Topological Manifolds

Nishtha Tikalal

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1 Motivation

Topological spaces are so numerous that mathematicians cannot classify them. There are many topological notions (Hausdorff, connected, paracompact, compact, ...), some Boolean-valued, some group-valued. No known set of such notions decides whether two topological spaces are homeomorphic. Physically, most topological spaces also resemble what we think of as spacetime very little.

For classical spacetime physics we therefore focus on a special type of topological space: those that can be *charted*, analogously to how the surface of the Earth is charted by an atlas. These are the **topological manifolds**.

2 Topological manifolds

Definition 1 (Topological manifold)**.** A topological space $(M, \mathcal{O})$ is a *d-dimensional topological manifold* if for every point $p \in M$ there exists an open set $U \in \mathcal{O}$ with $p \in U$ and a map

$$x : U \to x(U) \subseteq \mathbb{R}^d$$

such that

(i) x is invertible (there is $x^{-1} : x(U) \to U$),

(ii) x is continuous,

(iii) x^{-1} is continuous.

Here U carries the subset topology inherited from $\mathcal{O}$, and $\mathbb{R}^d$ carries the standard topology.

Remark. Continuity of x in both directions requires explicit assumption, since a continuous bijection need not have a continuous inverse (see Lecture 1). Because x^{-1} is continuous and U is open, the image $x(U) = \mathrm{preim}_{x^{-1}}(U)$ is open in $\mathbb{R}^d$.

Remark. There is a lot of data in this definition: for every point one needs an open neighborhood *together with* a whole map. The topology $\mathcal{O}$ on M is arbitrary, not the standard topology (which is only defined on $\mathbb{R}^d$).

2.1 Examples

Example 1 (Torus)**.** Let M be the surface of a donut (or car tire), embedded in $\mathbb{R}^3$ and equipped with the subset topology from the standard topology of $\mathbb{R}^3$. Open sets in M are of the form (open set in $\mathbb{R}^3$) $\cap\, M$. Around any point one finds such an open set and a continuous invertible map with continuous inverse onto an open region of $\mathbb{R}^2$. So M is a two-dimensional topological manifold (checked here only informally).

Example 2 (Distorted figure eight / closed wire)**.** A wire (e.g. a figure eight whose strands do not actually touch) in $\mathbb{R}^3$ with the subset topology is a one-dimensional topological manifold: around each point there is a piece of the wire that maps bijectively and bicontinuously to an open piece of $\mathbb{R}$.

Example 3 (Branching curve: not a manifold)**.** Take a curve that bifurcates, as a subset $M \subset \mathbb{R}^2$ with the subset topology. This is a topological space, but not a topological manifold. Near the branch point one can find open sets, but there is no continuous invertible map with continuous inverse from such a neighborhood into $\mathbb{R}^1$ or $\mathbb{R}^2$ (this is argued intuitively). The definition demands a single d that works everywhere, and even allowing varying dimension would not solve what to do at the junction.

3 Charts, atlases, coordinates

3.1 Real world versus chart

Think of M as the *real world*, e.g. the foot of the Eiffel Tower is a physical point $p \in M$. A chart map x sends it to numbers $x(p) \in \mathbb{R}^d$ (say $(1, 2)$). This is like a page of an atlas. The numbers mean nothing physical by themselves: they depend on the publisher's choice of chart (where the origin of the page is, whether the paper is stretched, etc.). Charts are a figment of our imagination; the real world does not care about them.

Definition 2 (Chart)**.** A pair (U, x) with U and x as in the definition above is called a *chart* of the topological manifold $(M, \mathcal{O})$. U is the *chart domain* (e.g. the area of Paris in the real world); $x(U)$ is its image in the chart (Paris on paper). The definition requires that every point of M lies in the domain of some chart.

Definition 3 (Atlas)**.** A collection of charts $\{(U_\alpha, x_\alpha)\}_{\alpha \in A}$, with A an arbitrary index set, is an *atlas* of M if

$$\bigcup_{\alpha \in A} U_\alpha = M,$$

i.e. the chart domains (which may overlap heavily) cover all of M.

3.2 Coordinate maps and coordinates

Since $x(p) \in \mathbb{R}^d$ is a d-tuple of reals, a chart map is equivalent to d real-valued maps

$$x^i : U \to \mathbb{R}, \qquad i = 1, \ldots, d, \qquad x(p) = \big(x^1(p), \ldots, x^d(p)\big).$$

These are the **coordinate maps** (component maps) of the chart. For $p \in U$, the number $x^i(p)$ is the *i-th coordinate of p with respect to the chart* (U, x). Choosing different generalized coordinates (as in Lagrangian mechanics) is the same as choosing different charts.

Remark (Chart-independent physics)*.* A stone does not respond to our imagining one chart or another. Otherwise we would have "physics with telekinesis". Until experiment says otherwise, we do chart-independent physics: the foot of the Eiffel Tower is a point $p \in M$ that is not moved by changing coordinates. This is a key point; it took Einstein years to fully clarify it.

Remark (Sets as "powder")*.* A set has no order, so $\{1, 2\} = \{2, 1\}$. The pure set $\mathbb{R}^2$ is like a powder: you can shuffle it, destroying all relations between points. Equipped with the standard topology it is more like a rubber sheet, which may be stretched but not torn (tearing would be discontinuous).

3.3 Example: two charts on the same domain

Let $M = \mathbb{R}^2$ with the standard topology.

Chart 1. Let $U = \mathbb{R}^2 \setminus \{(0,0)\}$ and define

$$x : U \to \mathbb{R}^2, \qquad (m,n) \mapsto (-m,-n).$$

This is invertible and continuous in both directions, so it is a perfectly good chart, even though it "inverts" the picture.

Chart 2 (polar). On a slit plane $V = \mathbb{R}^2 \setminus \{(a,0) \mid a \in \mathbb{R}_0^+\}$ (removing the nonnegative x-axis including the origin) define

$$y : V \to \mathbb{R}^2, \qquad (m,n) \mapsto \big(r,\varphi\big), \quad r = \sqrt{m^2+n^2},\ \varphi \in (0,2\pi) \text{ the polar angle}$$

(the lecture writes φ schematically as $\arctan(n/m)$). The image $y(V)$ is a half-infinite strip. A circle about the origin in the real world appears as a segment (missing one point) in this chart, whereas in Chart 1 it still appears as a circle. Both charts represent the same real world situation.

4 Chart transition maps

Let (U,x) and (V,y) be two charts with $U \cap V \neq \emptyset$. A point $p \in U \cap V$ has two chart representatives, $x(p)$ and $y(p)$ (like the Eiffel Tower appearing on two pages of an atlas). Restricting both chart maps to $U \cap V$, we get subsets $x(U \cap V)$ and $y(U \cap V)$ of $\mathbb{R}^d$.

Definition 4 (Chart transition map)**.** The *chart transition map* from (U,x) to (V,y) is

$$y \circ x^{-1} : x(U \cap V) \to y(U \cap V).$$

It first goes back from chart coordinates to the real point via x^{-1} and then forward to the other chart coordinates via y. Physically it is what is called a "change of coordinates".

Since x^{-1} and y are continuous, $y \circ x^{-1}$ is a continuous map between open subsets of $\mathbb{R}^d$ (composition of continuous maps is continuous, Lecture 1). It is thus an ordinary undergraduate-analysis map $\mathbb{R}^d \supseteq \cdot \to \mathbb{R}^d$.

Gluing. Chart transition maps contain the instructions for gluing together the charts of an atlas. Cutting out the pages of an atlas and gluing them together at their overlaps, identifying points of different charts that represent the same real point, reconstructs the global shape of the manifold (for an atlas of the Earth you get a sphere). The global topological information is thus encoded in the transition maps.

5 Manifold philosophy

The idea: define properties of objects in the real world by pushing them down to charts and examining their chart representatives, where the mathematics is that of $\mathbb{R}^d$ and undergraduate analysis.

Example: continuity of a curve. Let $\gamma : \mathbb{R} \to M$ be a curve in the real world (the course of a ship). Take a chart (U,x) (restricting attention to the part of the curve in U). Its chart representative is

$$x \circ \gamma : \mathbb{R} \to x(U) \subseteq \mathbb{R}^d,$$

a curve in $\mathbb{R}^d$, which is what undergraduate physics calls the trajectory. One can say: γ is continuous if $x \circ \gamma$ is continuous (in the standard topologies).

Advantage. We can use ordinary analysis on $\mathbb{R}^d$.

Disadvantages.

1. Philosophical: one may confuse the chart image with the real object.
2. The definition may be **ill-defined**: it assigns a property to γ using a chart x, which is an arbitrary choice. We must ensure that the property does not change if another chart is used (like checking that an operation on equivalence classes is independent of the representative).

Chart independence of continuity. Suppose $x \circ \gamma$ is continuous and let (V, y) be another chart (on the relevant part of the curve). Then

$$y \circ \gamma = y \circ \underbrace{x^{-1} \circ x}_{\mathrm{id}} \circ \gamma = \big(y \circ x^{-1}\big) \circ \big(x \circ \gamma\big).$$

The first factor is a chart transition map, hence continuous; the second is continuous by assumption. So $y \circ \gamma$ is continuous: the definition is independent of the chart. The continuity of the chart transition maps makes the definition well-defined.

Differentiability is different. One cannot define differentiability of γ directly on M, since there is no addition or scalar multiplication there to form a difference quotient; only chart representatives $x \circ \gamma : \mathbb{R} \to \mathbb{R}^d$ can be tested. But the above argument fails: a differentiable map composed with a merely continuous transition map $y \circ x^{-1}$ can be non-differentiable (a smooth curve can acquire a corner and lose its unique tangent). Hence on a mere topological manifold, differentiability cannot be defined through charts in a chart-independent way.

Fix (preview of the next lectures). Restrict the atlas so that all transition maps are differentiable (smooth). Compositions of differentiable maps are differentiable, so differentiability becomes chart-independent *for charts from this restricted atlas.* Starting from a *maximal* atlas (one containing every possible topological chart), throwing away troublesome charts is additional information. Choosing which charts to keep is the prescription of a **differentiable (smooth) atlas**, which is the content of the second next lecture and is a major step toward spacetime.

Remark (Question from the audience). Above, we only showed that continuity of $x \circ \gamma$ implies continuity of $y \circ \gamma$ ("lifted" continuity is chart-independent). That this chart-based continuity agrees with continuity of $\gamma : \mathbb{R} \to M$ as defined in Lecture 1 can be shown in a few lines; the lecturer notes it is somewhat of an accident of topology that continuity can be decided at the manifold level at all, which is why this point is de-emphasized. (Whether continuity of the inverse chart map is needed for one implication was left open in the lecture.)

Summary

- A **topological manifold** is a topological space such that every point lies in the domain of a **chart** (U, x): $x : U \to x(U) \subseteq \mathbb{R}^d$ is a homeomorphism onto an open subset of $\mathbb{R}^d$.
- An **atlas** is a collection of charts whose domains cover M. Coordinates $x^i(p)$ are the components of the chart map.
- Charts and coordinates are human inventions; the points of M are the real world. Physics should not depend on the choice of chart.

- **Chart transition maps** $y \circ x^{-1}$ are continuous maps between open subsets of $\mathbb{R}^d$ and encode how to glue the charts.
- **Manifold philosophy:** define properties of real-world objects via chart representatives, but check they are chart independent. Continuity is; differentiability requires extra structure (a differentiable atlas).